



THE PART NOBODY BENCHMARKS

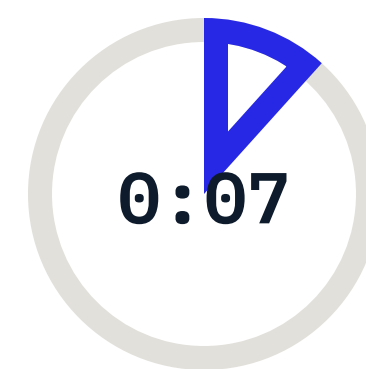
Harness.

`/'hɑːr.nɪs/` noun

Everything around the model: tools, loop, context, memory, sandbox, verification, guardrails.

The model is the part you choose. The harness is the part you build.

ONE IDEA · 60 SECONDS · EVERY WEEK



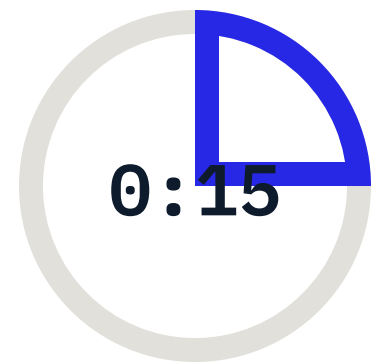
THE DEFINITION

**Agent = Model
+ Harness.**

The model is the brain — it reasons and decides.
The harness is the body and the workspace: it acts, remembers, verifies, and enforces limits.

“If you're not the model, you're the harness.”

VIVEK TRIVEDY, LANGCHAIN



HOW IT RUNS

MODEL

decides an action

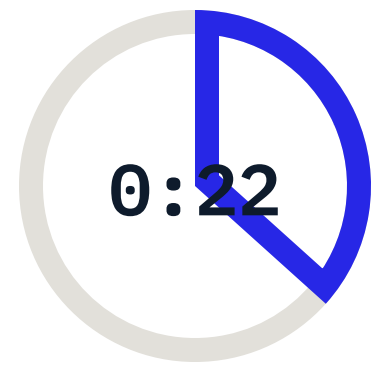


HARNESS

executes the tool, captures the result,
and updates context

🔄 **repeat until done or stopped**

The ReAct loop (Yao et al., 2022). The complexity isn't in the loop — it's in everything the loop manages.



THE PROGRESSION

1 PROMPT ENGINEERING

wording the ask



2 CONTEXT ENGINEERING

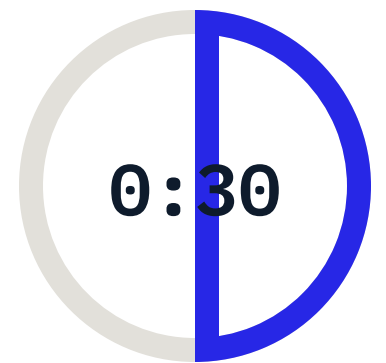
curating what the model sees



3 HARNESS ENGINEERING

designing the whole system around it

Each contains the last. Prompts and context live inside the harness.



THE BUILDING BLOCKS

SYSTEM PROMPT

standing instructions and constraints

TOOLS + EXECUTION

model decides; harness executes

SANDBOX + FILES

workspace where the agent operates

MEMORY + COMPACTION

what persists; what gets summarized

VERIFICATION

test, inspect, validate, correct course

GUARDRAILS + OBSERVABILITY

approvals, limits, logs, full trace



THE EVIDENCE

Researchers fixed the model, then ran the same coding tasks through five different harnesses.

Same model. Five harnesses.

GLM 5.1 · 12.5 points

60.9% → 73.4%

Qwen 3.6-flash · 27.4 points

38.6% → 66.0%

Only the harness changed. A separate study — 106 tasks, 5,194 runs — reached the same conclusion.

SOURCES: CLAW-SWE-BENCH · HARNESS-BENCH (ARXIV, 2026) ·
PASS@1 = SOLVED ON FIRST ATTEMPT



WHERE HARNESSSES BREAK

CONTEXT ROT

history grows, reasoning degrades

TOOL OVERLOAD

too many options, worse decisions

BRITTLE WIRING

a reworded description breaks the call

WEAK VERIFICATION

success declared on incomplete work

MISSING GUARDRAILS

irreversible action, no approval

These aren't necessarily model problems. The model gets the blame anyway.

TIME'S UP

**Model choice is a decision.
Harness design is a
discipline.**

START DUMB – one reliable loop first

WRITE TOOL DESCRIPTIONS LIKE DOCS

INSTRUMENT EVERYTHING – you can't debug
what you can't see

Save this for your next agent build.